

HDOC9

Question 1:

Write a menu-driven C program to input two numbers and swap them using a third variable & without using third variable. Also display the values before and after swapping.

Understanding the Problem Statement

1. Input

The program takes:

- Two variables as input

Required Output

The program should Swap the values of the two variables that were given

Formula Used:

$$A=a+b$$

$$B=a-b$$

$$A=a-b$$

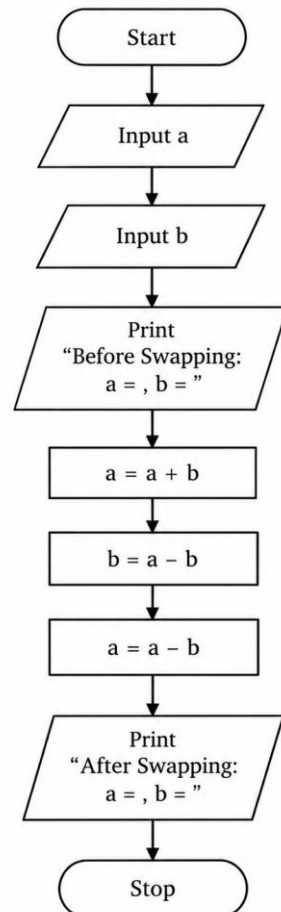
2. Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	A=10,B=12	A=10,B=12	A=10,B=12	Pass
2	A=15,B=20	A=15,B=20	A=15,B=20	Pass
3	A=100,B=140	A=100,B=140	A=100,B=140	Pass
4	A=150,B=200	A=150,B=200	A=150,B=200	Pass
5	A=5,B=10	A=5,B=10	A=5,B=10	Pass

3. Evaluation of the Algorithm

The algorithm takes input for two variables from the user, then the program code swaps the value of the variable. The algorithm is

4. FLOWCHART



5. C- Program

```
#include <stdio.h>
```

```
int main() {
```

```
    int a, b;
```

```
    printf("Enter the first number (a): ");
```

```
    scanf("%d", &a);
```

```
    printf("Enter the second number (b): ");
```

```
    scanf("%d", &b);
```

```
    printf("\nBefore Swapping: a = %d, b =
```

```

%d\n", a, b);

a = a + b;

b = a - b;

a = a - b;

printf("After Swapping: a = %d, b = %d\n", a,
b);

return 0;

}

```

Save the program with the filename:

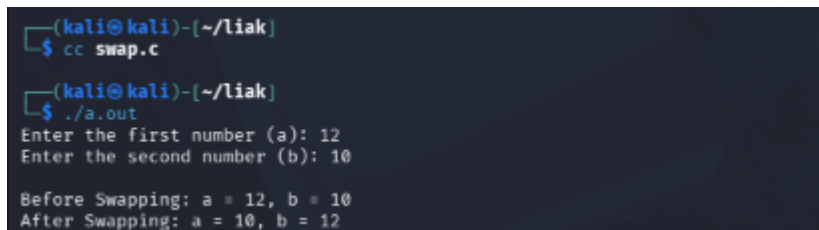
swap.c

Compile it using CC:

cc vol.c

If there are no errors, execute the program using:

./a.out



```

(kali@kali)~$ cc swap.c
(kali@kali)~$ ./a.out
Enter the first number (a): 12
Enter the second number (b): 10
Before Swapping: a = 12, b = 10
After Swapping: a = 10, b = 12

```

6. Sample Execution:

Test Case:

Enter the first number (a): 12

Enter the second number (b): 10

Before Swapping: a=12 b=10

After Swapping: a=10 b=12

7. Evaluation against test cases:

The program was tested using different values of a and b including large and small values. In each case the output was coming to be correct when verified manually.

Therefore, All the test cases were successfully passed.

8. RESULT

Thus, the C program to swap the value of two variables without using a third variable was successfully written, compiled and executed. The program was tested with multiple test cases and produced expected results.

```
(kali@kali)-[~/liak]
$ cc swap.c

(kali@kali)-[~/liak]
$ ./a.out
Enter the first number (a): 12
Enter the second number (b): 10

Before Swapping: a = 12, b = 10
After Swapping: a = 10, b = 12
```

Question 2:

Write a **menu-driven** C program to find the sum of the first n natural numbers and sum of squares of the first n natural numbers.

Understanding the Problem Statement

1. Input

The program gets input for the choice, thus goes forward with the given input accordingly.

Required Output

The program should calculate:

Either the sum of first n natural numbers
or sum of squares of first n natural
numbers

Formula Used:

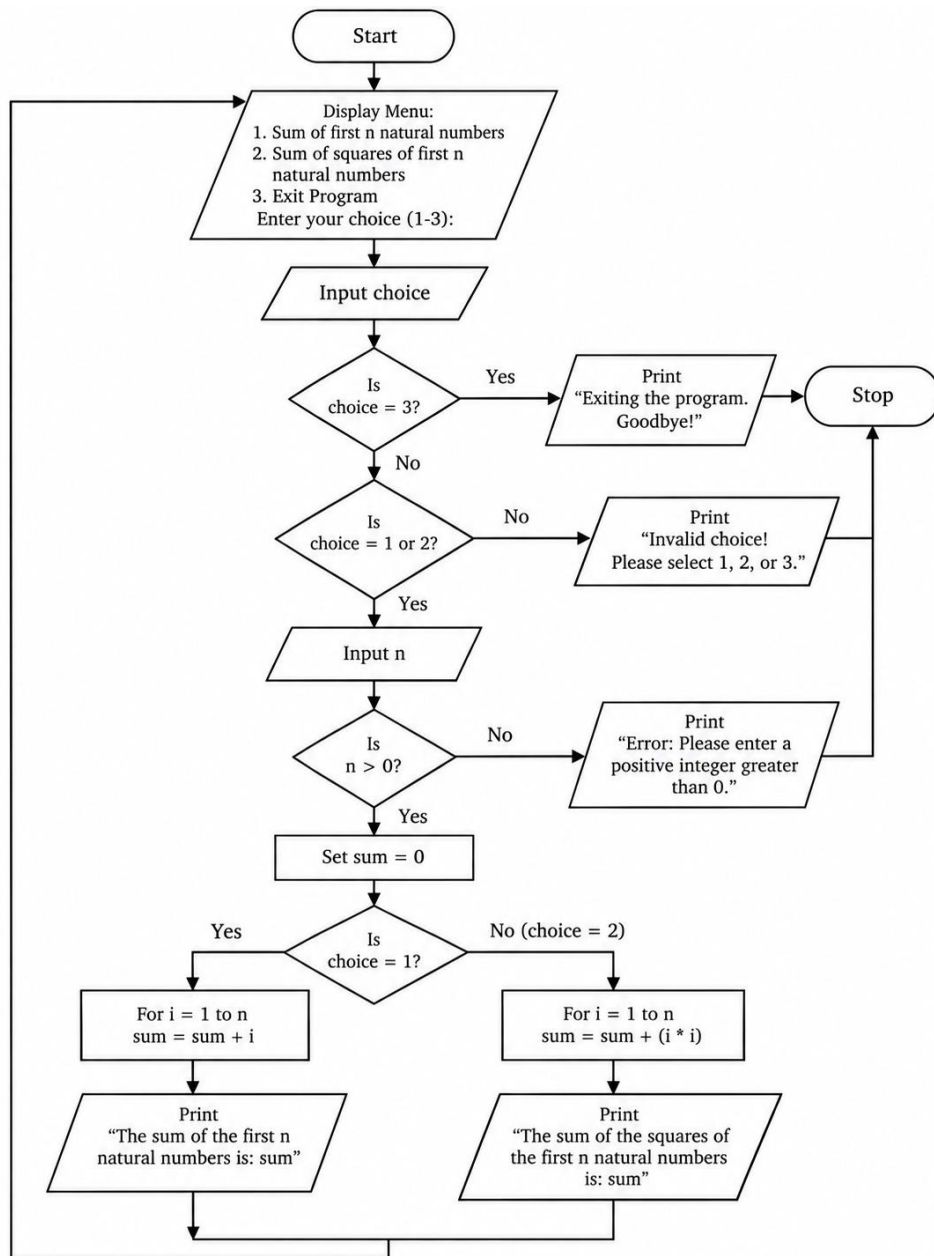
Sum of n natural numbers: inside the loop (sum =sum+i), i goes upto n

Sum of square of n natural numbers: inside loop (sum=sum+i*i), i goes upto n

2. Evaluation of the Algorithm

The algorithm takes initial input and gives three options to the user to get, 1. To get the sum of first n natural numbers, 2. To get sum of squares of first n natural numbers and the last one 3. Is to exit the program... then it asks for the input of n... It uses the given formula and finally gives the output.

3. Flowchart:



4. C- Program

```
#include <stdio.h>
```

```
int main() {
```

```
    int choice, n;
```

```
    int sum;
```

```
    do {
```

```
        printf("\n--- MATHEMATICS MENU ---\n");
```

```
        printf("1. Find the sum of first 'n' natural\nnumbers\n");
```

```
        printf("2. Find the sum of squares of first\n'n' natural numbers\n");
```

```
        printf("3. Exit Program\n");
```

```
        printf("Enter your choice (1-3): ");
```

```
        scanf("%d", &choice);
```

```
        if (choice == 3) {
```

```
            printf("Exiting the program.\nGoodbye!\n");
```

```
            return 0;
```

```
        }
```

```
        if (choice == 1 || choice == 2) {
```

```
            printf("Enter a positive integer 'n': ");
```

```
            scanf("%d", &n);
```

```
            if (n <= 0) {
```

```
                printf("Error: Please enter a positive\ninteger greater than 0.\n");
```

```
                continue;
```

```
            }
```

```
            sum = 0;
```

```
            switch (choice) {
```

```

        case 1:
            for (int i = 1; i <= n; i++) {
                sum = sum + i;
            }
            printf("The sum of the first %d
natural numbers is: %d\n", n, sum);
            break;

        case 2:
            for (int i = 1; i <= n; i++) {
                sum = sum + (i * i);
            }
            printf("The sum of the squares of
the first %d natural numbers is: %d\n", n,
sum);
            break;
        }
    } else {
        printf("Invalid choice! Please select 1,
2, or 3.\n");
    }

} while (choice != 3);

return 0;
}

```

Save the program with the filename:

gedit menu.c

Compile it using CC:

cc menu.c

If there are no errors, execute the program using:

`./a.out`

```
(kali㉿kali)-[~/liak]
└─$ cc menu.c

(kali㉿kali)-[~/liak]
└─$ ./a.out

— MATHEMATICS MENU —
1. Find the sum of first 'n' natural numbers
2. Find the sum of squares of first 'n' natural numbers
3. Exit Program
Enter your choice (1-3): 1
Enter a positive integer 'n': 3
The sum of the first 3 natural numbers is: 6

— MATHEMATICS MENU —
1. Find the sum of first 'n' natural numbers
2. Find the sum of squares of first 'n' natural numbers
3. Exit Program
Enter your choice (1-3): 2
Enter a positive integer 'n': 3
The sum of the squares of the first 3 natural numbers is: 14

— MATHEMATICS MENU —
1. Find the sum of first 'n' natural numbers
2. Find the sum of squares of first 'n' natural numbers
3. Exit Program
Enter your choice (1-3): 3
Exiting the program. Goodbye!
```

5. Sample Execution:

Test Case:

Enter your choice: 2

Enter a positive integer: 3

The sum of the first 3 natural numbers is: 14

6. Evaluation against test cases:

The program was tested using different value of n including large and small values. In each case the required sum was coming to be correct when verified manually.

Therefore, All the test cases were successfully passed.

7. RESULT

Thus, the C program to make a menu driven program to find sum of first n natural numbers or sum of squares of first n natural numbers was successfully written, compiled and executed. The program was tested with multiple test cases and produced expected results.


```
(kali@kali)-[~/liak]
$ cc menu.c

(kali@kali)-[~/liak]
$ ./a.out

— MATHEMATICS MENU —
1. Find the sum of first 'n' natural numbers
2. Find the sum of squares of first 'n' natural numbers
3. Exit Program
Enter your choice (1-3): 1
Enter a positive integer 'n': 3
The sum of the first 3 natural numbers is: 6

— MATHEMATICS MENU —
1. Find the sum of first 'n' natural numbers
2. Find the sum of squares of first 'n' natural numbers
3. Exit Program
Enter your choice (1-3): 2
Enter a positive integer 'n': 3
The sum of the squares of the first 3 natural numbers is: 14

— MATHEMATICS MENU —
1. Find the sum of first 'n' natural numbers
2. Find the sum of squares of first 'n' natural numbers
3. Exit Program
Enter your choice (1-3): 3
Exiting the program. Goodbye!
```

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